

Auto Profit Hub

Project Working, Workflow, and Feature Reference

Vehicle inventory, document registry, sales, customer, and profitability management system

Purpose: Auto Profit Hub helps a dealership track vehicles from purchase through repair, advertising, sale, customer follow-up, reporting, and document compliance.

1. Executive Summary

The project is a full-stack vehicle inventory and dealership operations platform. It combines a React frontend, Express API, Prisma data layer, MongoDB storage, protected user authentication, multi-tenant dealership data separation, and document-processing tools for purchase and sale records.

- Centralizes inventory, sales, expenses, repairs, advertising, customer data, reports, and used vehicle records.
- Supports staff, manager, admin, and platform admin workflows with role-aware screens.
- Automates parts of the paperwork process by extracting vehicle and sale data from documents.
- Calculates profitability from sale price, purchase cost, repair cost, and operating inputs.
- Keeps customer records connected to sold vehicles, visitors, leads, and follow-up categories.

2. Main System Workflow

1. User signs in and the app loads dealership-specific data.
2. Vehicle is added manually or from a purchase/source document.
3. Inventory stores VIN, make, model, year, mileage, color, title number, purchase cost, seller details, and document availability.
4. Repairs, advertising, and business expenses are recorded and can be linked to vehicles where applicable.
5. Bill of Sale is uploaded or sale is entered manually when the vehicle is sold.
6. Sale record is created, vehicle status changes to Sold, profit is calculated, and the Used Vehicle Record is updated.
7. Customer details from the sale are stored in Customers; users can also add visitors, leads, follow-ups, and manually linked vehicles.
8. Dashboard, Sales, Reports, Cash Flow, and Team Analytics summarize performance.

3. Feature Map

Area	What It Does	Main Capabilities
Authentication	Protects dealership data and routes.	Login, registration, JWT sessions, role-based protected routes.

Area	What It Does	Main Capabilities
Dashboard	Shows the operational snapshot.	Inventory status, sales performance, profit summary, recent activity.
Inventory	Tracks every vehicle lifecycle.	Add/edit/delete vehicles, statuses, purchase data, documents, repairs, sale action.
Sales	Stores finalized sales.	Sale table/cards, revenue, profit, bill-of-sale documents, edit/delete sale records.
Customers	Manages buyers, visitors, and leads.	Manual add, edit contact, categories, linked inventory vehicles, search, import from sales.
Used Forms	Handles document-driven workflows.	Bill of Sale upload, repair bill upload, used vehicle form generation.
Registry	Maintains document compliance data.	Used Vehicle Record entries, acquisition/disposition details, document updates.
Repairs	Adds vehicle reconditioning cost.	Parts/labor cost, shop, repair description, vehicle profit impact.
Advertising	Tracks campaign spend.	Campaigns, platforms, dates, spend, optional vehicle link.
Expenses	Tracks operating costs.	Business expense categories, date, notes, edit/delete.
Reports	Exports and summarizes results.	Revenue report, sales breakdown, profit reporting.
Team	Admin visibility into staff activity.	Users, roles, sales activity, performance analytics.
Settings	Dealership profile management.	Name, address, phone, email, logo/profile information.
Super Admin	Platform-level management.	Dealership oversight and platform admin functions.
AI Assistant	Helps query operations.	Ask about inventory, sales, and profit context from within the app.

4. Inventory and Vehicle Management

Inventory is the starting point for the operational workflow. Each vehicle record stores key identification, purchase, cost, status, and document information.

- Vehicle fields include VIN, make, model, year, mileage, color, purchase date, title number, and status.
- Purchase data includes seller name/address, purchase price, buyer fee, transport, inspection, registration, payment method, and total purchase cost.
- Vehicle status can move through Available, Reserved, Sold, or Returned depending on business activity.
- Documents are tracked with flags for generated Used Vehicle Record, source document, and Bill of Sale.
- Vehicle detail dialog provides tabs for overview, repairs, advertising, sale processing, documents, and customer viewing notes.

5. Sales and Profit Workflow

Sales can be created manually from a vehicle detail screen or automatically when a Bill of Sale document is processed. The sale stores buyer information, sale date, sale price, payment method, and document archive data.

Profit formula	Sale price - purchase cost - repair cost.
Sale action	Marks the vehicle as Sold and creates or updates the sale record.
Document update	Regenerates the Used Vehicle Record with disposition details.
Customer update	Creates or updates a customer record with contact, category, and linked vehicle metadata.
Bill of Sale	Can be uploaded, previewed, downloaded, and stored as base64 document data.

6. Customers Section

The Customers section stores people connected to the dealership. It supports buyers imported from sales and manual customers such as visitors, leads, and follow-ups.

- Manual customer creation with first name, last name, phone, email, address, city, state, and zip.
- Customer categories: Bought Vehicle, Came for Visit, Lead, Follow Up, and Other.
- Vehicle linking from Inventory with search by VIN, make, model, year, or status.
- Main customer search across name, contact number, email, category, address, and linked vehicle.
- Sales import creates or updates customer records from existing sale data.
- Existing customers can be edited to update contact details, category, and vehicle link.

7. Document Processing and Registry

The system includes document workflows for dealership paperwork. Purchase/source documents and Bills of Sale can be processed to extract vehicle and transaction details.

- Document parser extracts VIN, year, make, model, title information, seller/source details, buyer/disposition details, dates, and prices.
- Used Vehicle Record PDF generation fills acquisition and disposition sections.

- Registry keeps VIN-based document records with source file name, document type, purchase details, and sale/disposition details.
- Document download and preview functions let users access records without manually searching file storage.
- Bulk import tooling can align purchase and sale documents to inventory records.

8. Financial and Operational Modules

Area	What It Does	Main Capabilities
Repairs	Captures vehicle-specific reconditioning costs.	Repair shop, parts cost, labor cost, repair date, description.
Advertising	Tracks marketing spend and optional vehicle association.	Campaign name, platform, start/end dates, amount spent, linked vehicle.
Business Expenses	Captures non-vehicle operating expenses.	Category, amount, date, notes.
Cash Flow	Summarizes money movement.	Income and cost visibility for managers/admins.
Reports	Provides management exports and summaries.	Revenue reports, unit counts, gross revenue, net profit, buyer breakdowns.

9. User Roles and Access

Staff	Operational access to inventory, sales, used forms, customers, and day-to-day records. Profit can be masked in some views.
Manager	Broader access to registry, reports, customers, and sales management functions.
Admin	Full dealership management including team analytics, settings, reports, customers, inventory, and operations.
Super Admin	Platform-level administration across dealerships.

10. Technical Architecture

Frontend	React, TypeScript, Vite, React Router, TanStack Query, Tailwind/shadcn-style UI components.
Backend	Node.js, Express routes, authentication middleware, tenant injection middleware, file upload handling.
Database	MongoDB through Prisma Client with models for dealership, users, vehicles, sales, purchases, repairs, customers, expenses, advertising, and registry.
Document services	Document parser and used vehicle PDF service for OCR/AI extraction and generated paperwork.
Caching	In-memory route cache for selected list endpoints such as vehicles and sales.
Security	JWT authentication, role-aware protected routes, helmet, CORS, rate limiting, and dealership isolation.

11. Key Data Models

Area	What It Does	Main Capabilities
Dealership	Tenant container.	Users, vehicles, sales, purchases, repairs, customers, documents, expenses.
User	Authenticated app user.	Email, password, name, role, dealership relation.
Vehicle	Inventory item.	VIN, make/model/year, mileage, status, title number, purchase, sale, repairs.
Purchase	Acquisition record.	Seller, purchase price, fees, total cost, document data.
Sale	Disposition record.	Customer name, phone, address, sale date, sale price, payment, profit, Bill of Sale.
Customer	Buyer/visitor/lead.	Name, email, phone, address, license, notes metadata.
DocumentRegistry	Compliance document record.	Vehicle identity, acquisition and disposition fields, document base64.
Repair	Reconditioning record.	Vehicle, shop, parts, labor, date, description.
Expense/Advertising	Cost tracking.	Business cost and campaign spend records.

12. Recent Customer Enhancements

- Customers route is mounted under /api/customers.
- Customers page is available in desktop sidebar and mobile navigation.
- Sales and Bill of Sale flows create or update customers automatically.
- Manual Add Customer workflow supports categories and inventory vehicle linking.
- Customer import from sales updates existing customer records instead of only creating new ones.
- Customer categories and vehicle links are stored in the existing notes field to avoid requiring an immediate database migration.

13. Operational Notes

- Restart the backend server after backend route changes.
- Run npm run build after frontend changes to verify TypeScript and Vite production build.
- If Prisma schema changes are made, regenerate Prisma Client before running the server.
- Avoid pushing local changes to GitHub unless explicitly requested.
- Keep customer contact data accurate because it is reused across sales records and customer follow-up workflows.

14. End-to-End Example

1. Add a vehicle to Inventory from purchase details or source document.
2. Attach repair bills and advertising spend as work is completed.
3. Upload a Bill of Sale or manually process the sale from the vehicle record.

4. The system marks the vehicle Sold, calculates profit, updates the Used Vehicle Record, and saves the Bill of Sale.
5. The buyer appears in Customers as Bought Vehicle and can be edited with email, phone, category, and vehicle link.
6. Managers review Sales, Reports, Cash Flow, and Dashboard totals.